Binbin Wang, Ph.D.

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Professional Summary

- **Computational biologist** with **11** years of experience in **bioinformatics** and **systems biology**, specializing in **precision oncology** and **immune checkpoint blockade** (ICB) response prediction
- Expertise in **large-scale omics data analysis**, **algorithm development**, and **machine learning** applications in cancer research
- Published 16 papers in high-impact journals, including four first-author publications including ***Cell***, ***Nature Protocols***, ***Genomics, Proteomics & Bioinformatics***, and ***iScience***
- Created multiple widely used computational frameworks to identify drug combinations for targeted therapy and immune checkpoint therapy

Technical Skills

- **Bioinformatics & Computational Biology**: CRISPR screen, single-cell RNA-seq, RNA-seq, ChIP-seq, ATAC-seq, DNA-seq, mass spectrometry data analysis
- **Programming**: Python, R, Shell, HTML
- **Machine Learning & Data Analysis**: Machine Learning, Statistical modeling, high-performance computing
- **Algorithm Development**: MAGeCKFlute, LiBIO, DETACH
- **Data Processing & Pipeline Development**: Snakemake, workflow automation, HPC computing
- **Code Version Control**: Git, Bitbucket

Research Experience

Postdoctoral Fellow, Cancer Data Science Laboratory, NCI (03/2021 - Present)
Advisor: Dr. Eytan Ruppin

- Developed **LiBIO**, an early-on-treatment transcriptional blood biomarker surpassing existing ICB response predictors in HNSCC (Manuscript under review).
- Identified decoupling genes (**DETACH**) influencing cytotoxic and exhausted T cell activities to enhance melanoma immunotherapy predictions (Published on ***iSciene***).
- Characterized bispecific antibody effectiveness and toxicity in cancer therapy (Manuscript in preparation).

Research Assistant, Shanghai Jiaotong University, Medical School (06/2019 - 03/2021)
Advisor: Dr. Leng Siew Yeap

- Developed an innovative algorithm for DNA sequencing analysis to characterize immunoglobulin gene diversification (Published on ***Science immunology***, citation **13**).
- Characterized antibody somatic hypermutation mechanisms and non-coding regulatory functions (Published on ***Cell***, citation **23**).

Graduate Research, Dana-Farber Cancer Institute & Tongji University (2014-2019)
Advisor: Dr. Xiaole Shirley Liu

- Developed **MAGeCKFlute**, a widely used computational pipeline for CRISPR screen data analysis (Published on ***Nature Protocols***, citation **367**).
- Identified therapeutic targets by integrating CRISPR screen and RNA-seq data analysis for BRAFi-resistant melanoma (Published on ***Genomics, Proteomics & Bioinformatics***, citation **21**).
- Designed and executed the **MusCK murine CRISPR knockout library**, conducting in vivo screens in TNBC models to identify key genetic regulators of tumor progression and response to PD-1 blockade (Published on ***Cell***, citation **135**).
- Contributed to the development of **TRUST** (Published on ***Nature genetics***, citation **80**), a novel computational method for de novo assembly of CDR3 regions from RNA-seq data, and **TIMER** (Published on ***Cancer research***, citation **5048**), a tool for characterizing tumor-immune interactions and immune infiltrates.

Education

Ph.D., Bioinformatics, Tongji University, Shanghai, China (2014-2019)
Visiting Scholar, Dana-Farber Cancer Institute, Harvard Medical School, Boston, USA (2017-2018)
B.A., Biotechnology, Northeast Agriculture University, Harbin, China (2008-2012)

Publications

1. **Binbin Wang***, Robert Saddawi-Konefka*, et al., Kun Wang, J. Silvio Gutkind, Eytan Ruppin, *An early on-treatment transcriptional blood biomarker best predicts immune checkpoint response of head and neck cancer patients.* (Submitting)
2. **Binbin Wang*,** Kun Wang, Peng Jiang, Eytan Ruppin*. Decoupling the correlation between cytotoxic and exhausted T lymphocyte states enhances melanoma immunotherapy response prediction.* **iScience, 2024**
3. **Binbin Wang***, Mei Wang*, Wubing Zhang*, et al., Wei Li#, Xiaole Shirley Liu#. *Integrative analysis of pooled CRISPR genetic screens using MAGeCKFlute.* **Nature Protocol, 2019**
4. Xiaoqing Wang*, Shengqing Gu*, Collin Tokheim*, **Binbin Wang***, et al., Myles Brown, X. Shirley Liu. *In vivo CRISPR screens identify the E3 ligase Cop1 as a modulator of macrophage infiltration and cancer immunotherapy target.* **Cell, 2021**
5. Ziyi Li*, **Binbin Wang***, et al., X. Shirley Liu. *CRISPR Screens Identify Essential Cell Growth Mediators in BRAF Inhibitor-resistant Melanoma.* **Genomics Proteomics Bioinformatics, 2020**
6. Sahil Sahni, **Binbin Wang,** et al., Kun Wang, Eytan Ruppin. *A machine learning model reveals expansive downregulation of ligand-receptor interactions that enhance lymphocyte infiltration in melanoma with developed resistance to immune checkpoint blockade.* **Nature Communications, 2024**
7. Qian Hao*, Chuanzong Zhan*, Chaoyang Lian, Simin Luo, Wenyi Cao, **Binbin Wang**, et al., Leng-Siew Yeap. *DNA repair mechanisms that promote insertion-deletion events*

during immunoglobulin gene diversification. **Science Immunology, 2023**

8. Yanyan Wang*, Senxin Zhang*, Xinrui Yang, Joyce K Hwang, Chuanzong Zhan, Chaoyang Lian, Chong Wang, Tuantuan Gui, **Binbin Wang**, et al., Leng-Siew Yeap, Fei-Long Meng. *Mesoscale DNA feature in antibody-coding sequence facilitates somatic hypermutation.* **Cell, 2023**
9. Collin Tokheim, Xiaoqing Wang, Richard T. Timms, Boning Zhang, Elijah L. Mena, **Binbin Wang**, et al., X. Shirley Liu. *Systematic characterization of mutations altering protein degradation in human cancers*. **Molecular Cell, 2021**
10. Shaokun Shu, Hua-Jun Wu, Jennifer Y. Ge, Rhamy Zeid, Isaac S. Harris, Bojana Jovanovic, Katherine Murphy, **Binbin Wang**, et al., Kornelia Polyak. *Synthetic lethal and resistance interactions with BET bromodomain inhibitors in triple-negative breast cancer.* **Molecular Cell, 2020**
11. Dongqing Sun, Jin Wang, Ya Han, Xin Dong, Jun Ge, Rongbin Zheng, Xiaoying Shi, **Binbin Wang**, Ziyi Li, et al., Chenfei Wang. *TISCH: a comprehensive web resource enabling interactive single-cell transcriptome visualization of tumor microenvironment*, **Nucleic Acids Research, 2020**
12. Xihao Hu, Jian Zhang, Jin Wang, Jingxin Fu, Taiwen Li, Xiaoqi Zheng, **Binbin Wang**, X. Shirley Liu. *Landscape of B cell immunity and related immune evasion in human cancers*. **Nature Genetics, 2019**
13. Jian Zhang, Xihao Hu, Jin Wang, Avinash Das Sahu, David Cohen, Li Song, Zhangyi Ouyang, Jingyu Fan, **Binbin Wang**, X Shirley Liu, *Immune receptor repertoires in pediatric and adult acute myeloid leukemia*, **Genome medicine, 2019**
14. Shenglin Mei, Clifford A Meyer, Rongbin Zheng, Qian Qin, Qiu Wu, Peng Jiang, Bo Li, Xiaohui Shi, **Binbin Wang**, Liu XS. *Cistrome Cancer: a web resource for integrative gene regulation modeling in cancer*, **Cancer research, 2017**
15. Bo Li, Taiwen Li, **Binbin Wang**, et al. X Shirley Liu. *Ultrasensitive detection of TCR hypervariable-region sequences in solid-tissue RNA-seq data*. **Nature Genetics, 2017**
16. Taiwen Li, Jingyu Fan, **Binbin Wang**, et al., X. Shirley Liu. *TIMER: A web server for comprehensive analysis of tumor-infiltrating immune cells*. **Cancer research, 2017**
17. Bo Li*, Taiwen Li*, Jean-Christophe Pignon, **Binbin Wang**, X. Shirley Liu. *Landscape of tumor-infiltrating T cell repertoire of human cancers*. **Nature Genetics, 2016**

Peer Review

Independently contributed to the review process for journals like *PLOS Computational Biology*, *Frontiers in Genetics*, *Cellular Physiology* and *Biochemistry*, *MDPI*, and *Bioinformatics*. Collaborated closely with my supervisor to review papers for esteemed journals such as *Cell*, *Cancer Discovery*, and *Genome Medicine*, *Science Translational Medicine*, *Patterns*.

Professional Memberships

- Reviewer for **PLOS Computational Biology**, **Frontiers in Genetics**, **Cellular Physiology and Biochemistry**, **MDPI,** and **Bioinformatics.**
- Scientific Poster Judge, NIH Postbac Poster Day 2024

Conferences & Presentations

- **AACR Annual Meeting 2024** – Poster presentation on bispecific antibodies in cancer treatment
- **AACR Annual Meeting 2023** – Poster presentation on ICB biomarkers
- **Cell Symposia 2022** – Poster presentation on T cell exhaustion and cytotoxicity decoupling
- **NCI Cancer Data Science Laboratory Seminar 2022** – Oral presentation on melanoma immunotherapy response prediction
- **Center for Functional Cancer Epigenetics Seminar 2017** – Oral presentation on synthetic lethal interactions in TNBC
- **Scientific and Technical Advances in Cancer Immunology 2019**
- **Computational analysis of single-cell data and cell-type heterogeneity: the next frontier 2018**
- **Mathematics, computer and life sciences cross research young scholars Forum 2018**
- **Systems Biology of Gene Regulation & Genome Editing Cold Spring Harbor Asia 2016**
- **The 7th Bioinformatics and systems biology conference 2016**
- **The 12th International Bioinformatics Workshop, IBW2015 2015**
- **Statistical Genomic Workshop 2015**

Awards & Honors

- National Scholarship for Doctoral Students (2018)
- Outstanding Student Cadres Award (2010)
- Student Award for Research and Innovation (2009)

Mentorship & Supervision

- Yaru Song, PhD candidate, Tongji University (2015 - 2016)
- Jingxin Fu, PhD candidate, Tongji University (2016 - 2017)
- Jin Wang, PhD candidate, Tongji University (2016 - 2017)
- Wubing Zhang, PhD candidate, Tongji University (2017 - 2018)
- Hantong, PhD candidate, Tongji University (2017 - 2018)
- Xiaoying Shi, PhD candidate, Tongji University (2018 - 2019)
- Dongqing Sun, PhD candidate, Tongji University (2018 - 2019)
- Ya Han, PhD candidate, Tongji University (2018 - 2019)
- LiangDong Sun, Master's student, Shanghai Pulmonary Hospital, Tongji University (2018 – 2021)
- Jun Ge, Master's student, Tongji University (2019 – 2021)

References

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Dr. Wei Li
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